



Travelling wave solutions for time-delayed nonlinear evolution equations

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ABSTRACT

Time-delayed nonlinear evolution equations have a wide range of applications in science and engineering. In this paper, the $\left(\frac{G'}{G}\right)$ -expansion method is implemented to establish travelling wave solutions for time-delayed Burgers and time-delayed Burgers–Fisher equations. The travelling wave solutions are expressed by hyperbolic functions and trigonometric functions. The results reveal that $\left(\frac{G'}{G}\right)$ -expansion method is very effective and a powerful tool for solving nonlinear time-delayed evolution equations arising in mathematical physics.

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1. Introduction

A number of nonlinear phenomena such as physical, biochemical and biological processes are described by the interplay of reaction and diffusion, or by the interaction between convection and diffusion. The well-known partial differential equation which governs a wide variety of these phenomena is the Burgers equation. The diffusion processes, however, get significantly modified when the memory effects are taken into account, i.e., when the dispersal of the particles is not mutually independent. This implies that the correlation between the successive movements of the diffusing particles may be understood as a delay in the flux for a given concentration gradient. Thus existence of time-delay is an important feature in convection–diffusion systems. It is interesting to point out that a well-known generalization of the Burgers–Fisher equation, namely the generalized time-delayed Burgers–Fisher equation is given by [1,2]

$$\tau u_{tt} + [1 - \tau f_u]u_t = u_{xx} - pu^s u_x + f(u), \quad f(u) = qu(1 - u^s), \quad (1)$$

where p, q, s are constants and τ is a time-delayed constant. Eq. (1) reduces to the classical Burgers equation when $q = \tau = 0$ and $p = s = 1$.

On the other hand, directly seeking exact solutions of nonlinear partial differential equations has become one of the central themes of perpetual interest in mathematical physics. In order to understand better the nonlinear phenomena as well as further application in practical life, it is important to seek more of their exact travelling wave solutions. Many methods are used to obtain travelling solitary wave solutions to nonlinear PDEs, such as the Adomian decomposition method [1,3], the sub-ODE method [4,5], F -expansion method [6,7], the homotopy perturbation method [8,9], the exp-function method [10–12], Hirota's bilinear method [13,14], Jacobi elliptic function method [15] and so on.

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However, practically there is no unified method that can be used to handle all types of nonlinear partial differential equations. Recently, Wang et al. [16] introduced a new direct method called the $\left(\frac{G'}{G}\right)$ -expansion method to look for travelling wave solutions of nonlinear evolution equations. One of the most effective straightforward method to construct exact solutions of PDEs is the $\left(\frac{G'}{G}\right)$ -expansion method [17–19]. The main ideas of the $\left(\frac{G'}{G}\right)$ -expansion method are that the travelling wave solutions of a nonlinear evolution equation can be expressed by a polynomial in $\left(\frac{G'}{G}\right)$, where $G = G(\xi)$ satisfies a second order linear ordinary differential equation (LODE) as follows: $G'' + \lambda G' + \mu G = 0$, $G' = \frac{dG(\xi)}{d\xi}$, $\xi = x - \omega t$, the degree of the polynomial can be determined by considering the homogeneous balance between the highest order derivatives and nonlinear terms appearing in a given nonlinear evolution equation, and the coefficients of the polynomial can be obtained by solving a set of algebraic equations resulted from the process of using the $\left(\frac{G'}{G}\right)$ -expansion method. Later Zhang et al. [20] proposed a generalized $\left(\frac{G'}{G}\right)$ -expansion method for solving variable-coefficient equations and high-dimensional equations.

In the present work, $\left(\frac{G'}{G}\right)$ -expansion method is implemented to obtain travelling wave solutions of the following time-delayed Burgers equation of the form

$$\tau u_{tt} + u_t = -pu^s u_x + u_{xx}, \quad (2)$$

for different values of τ , p , s and time-delayed Burgers–Fisher equations. Eq. (2) have been applied to forest fire [21], population growth models, Neolithic transitions [22,23] and many other areas of applied sciences.

2. Summary of the $\left(\frac{G'}{G}\right)$ -expansion method

A partial differential equation (PDE)

$$P(u, u_t, u_x, u_{xx}, u_{tt}, \dots) = 0, \quad (3)$$

can be converted to ordinary differential equation (ODE)

$$Q(u, -k\omega u', ku', k^2 u'', k^2 \omega^2 u'', \dots) = 0 \quad (4)$$

by using a wave variable $\xi = k(x - \omega t)$.

By $\left(\frac{G'}{G}\right)$ -expansion method the solution of ODE (4) can be expressed by a polynomial in $\left(\frac{G'}{G}\right)$ as follows [16]:

$$u(\xi) = \sum_{i=0}^m a_i \left(\frac{G'}{G}\right)^i, \quad (5)$$

where $G = G(\xi)$ satisfies the second order LODE in the form

$$G'' + \lambda G' + \mu G = 0, \quad (6)$$

here $a_1, \dots, a_m, \lambda$ and μ are constants to be determined later, $a_m \neq 0$. The positive integer m can be determined by considering the homogeneous balance between the highest order derivatives and nonlinear terms appearing in ODE (4).

By substituting Eq. (5) in Eq. (4) and using second order LODE (6), collecting all terms with the same order of $\left(\frac{G'}{G}\right)$ together, the left-hand side of Eq. (4) is converted into another polynomial in $\left(\frac{G'}{G}\right)$. Equating each coefficient of this polynomial to zero, yields a set of algebraic equations for $a_1, \dots, a_m, \lambda, k, \omega$ and μ . Solving this set of algebraic equations we can obtain the explicit expressions for $a_1, \dots, a_m, \lambda, k, \omega$ and μ . Substituting this values and general solutions of (6) in (5), we can obtain the exact solutions of (3).

In the remaining sections we will find the travelling wave solutions to nonlinear time-delayed evolution equations using $\left(\frac{G'}{G}\right)$ -expansion method.

3. Solutions to the time-delayed Burgers equation

To obtain the solution for Eq. (2), we consider the transformation $u = U(\xi)$, $\xi = k(x - \omega t)$, where k and ω are constants to be determined later. We can rewrite time-delayed Burgers equation (2) in the following nonlinear ordinary differential equation of the form

$$(\tau\omega^2 - 1)k^2 U'' - k\omega U' + pkU^s U' = 0. \quad (7)$$

Balancing $U^s U'$ with U'' in Eq. (7) gives $m = 1/s$. To get a closed form solution we use the transformation $U = v^{1/s}$ to change

Eq. (7) into

$$(\tau\omega^2 - 1)k^2 \left[v''v + \left(\frac{1}{s} - 1 \right) v'^2 \right] - kwvv' + pkv^2v' = 0. \quad (8)$$

Suppose that the solution of ODE (8) can be expressed by a polynomial in $\left(\frac{G'}{G}\right)$ as follows:

$$v(\xi) = \sum_{i=0}^m a_i \left(\frac{G'}{G} \right)^i, \quad (9)$$

where $G = G(\xi)$ satisfies the second order LODE in the form

$$G'' + \lambda G' + \mu G = 0. \quad (10)$$

By using Eqs. (9) and (10) it is derived that

$$v^2 = a_m^2 \left(\frac{G'}{G} \right)^{2m} + \dots, \quad (11)$$

$$v' = -ma_m \left(\frac{G'}{G} \right)^{m+1} + \dots, \quad (12)$$

$$v'' = m(m+1)a_m \left(\frac{G'}{G} \right)^{m+2} + \dots. \quad (13)$$

To determine the parameter m , we balance the linear terms of highest order in Eq. (8) with the highest order nonlinear terms. Considering the homogeneous balance between v'' and v^2 in (8), based on (11) and (13) we obtain that $2m + 2 = 2m + m + 1 \Rightarrow m = 1$, so we can write (9) as

$$v(\xi) = a_1 \left(\frac{G'}{G} \right) + a_0, \quad a_1 \neq 0, \quad (14)$$

By using Eqs. (9) and (10) it is derived that

$$v^2 = a_1^2 \left(\frac{G'}{G} \right)^2 + 2a_1a_0 \left(\frac{G'}{G} \right) + a_0^2, \quad (15)$$

$$v' = -a_1\mu - a_1\lambda \left(\frac{G'}{G} \right) - a_1 \left(\frac{G'}{G} \right)^2, \quad (16)$$

$$v'' = 2a_1 \left(\frac{G'}{G} \right)^3 + 3a_1\lambda \left(\frac{G'}{G} \right)^2 + (2a_1\mu + a_1\lambda^2) \left(\frac{G'}{G} \right) + a_1\lambda\mu. \quad (17)$$

By substituting (14)–(17) in (8) and collecting all terms with the same power of $\left(\frac{G'}{G}\right)$ together, the left-hand side of (8) is converted into another polynomial in $\left(\frac{G'}{G}\right)$. Equating each coefficient of this polynomial to zero, yields a set of simultaneous algebraic equations for a_1 , a_0 , ω and k as follows:

$$\begin{aligned} \left(\frac{G'}{G}\right)^0: & k^2 a_1^2 \mu^2 + k\omega a_0 a_1 \mu - pka_0^2 a_1 \mu + \frac{k^2 \tau \omega^2 a_1^2 \mu^2}{s} - k^2 a_1 \lambda \mu a_0 - \frac{k^2 a_1^2 \mu^2}{s} + k^2 \tau \omega^2 a_1 \lambda \mu a_0 - k^2 \tau \omega^2 a_1^2 \mu^2 = 0 \\ \left(\frac{G'}{G}\right)^1: & k\omega a_1^2 \mu + k\omega a_0 a_1 \lambda - 2pka_0 a_1^2 \mu - pka_0^2 a_1 \lambda - \frac{2k^2 a_1^2 \lambda \mu}{s} + \frac{2k^2 \tau \omega^2 a_1^2 \lambda \mu}{s} + 2k^2 \tau \omega^2 a_1 \mu a_0 - k^2 \tau \omega^2 a_1^2 \lambda \mu + k^2 a_1^2 \lambda \mu - k^2 a_1 \lambda^2 a_0 + \\ & k^2 \tau \omega^2 a_1 \lambda^2 a_0 - 2k^2 a_1 \mu a_0 = 0 \\ \left(\frac{G'}{G}\right)^2: & k\omega a_1^2 \lambda + k\omega a_0 a_1 - pka_1^3 \mu - pka_0^2 a_1 - 2pka_0 a_1^2 \lambda + \frac{2k^2 \tau \omega^2 a_1^2 \mu}{s} + \frac{k^2 \tau \omega^2 a_1^2 \lambda^2}{s} + 3k^2 \tau \omega^2 a_1 \lambda a_0 - 3k^2 a_1 \lambda a_0 - \frac{k^2 a_1^2 \lambda^2}{s} - \frac{2k^2 a_1^2 \mu}{s} = 0 \\ \left(\frac{G'}{G}\right)^3: & -pka_1^3 \lambda - 2pka_0 a_1^2 + 2k^2 \tau \omega^2 a_1 a_0 + \frac{2k^2 \tau \omega^2 a_1^2 \lambda}{s} - \frac{2k^2 a_1^2 \lambda}{s} + k^2 \tau \omega^2 a_1^2 \lambda - 2k^2 a_1 a_0 - k^2 a_1^2 \lambda + k\omega a_1^2 = 0 \\ \left(\frac{G'}{G}\right)^4: & -pka_1^3 + \frac{k^2 \tau \omega^2 a_1^2}{s} - k^2 a_1^2 - \frac{k^2 a_1^2}{s} + k^2 \tau \omega^2 a_1^2 = 0. \end{aligned}$$

Solving the above algebraic equations using Maple, we obtain the following set of nontrivial solutions

$$a_1 = \pm \frac{(s+1)\omega}{p\sqrt{\lambda^2 - 4\mu}}, \quad a_0 = \pm \frac{(s+1)\lambda\omega}{2p\sqrt{\lambda^2 - 4\mu}} + \frac{(s+1)\omega}{2p}, \quad k = \pm \frac{s\omega}{(\tau\omega^2 - 1)\sqrt{\lambda^2 - 4\mu}}, \quad (18)$$

where λ , μ , s , p , τ and ω are constants.

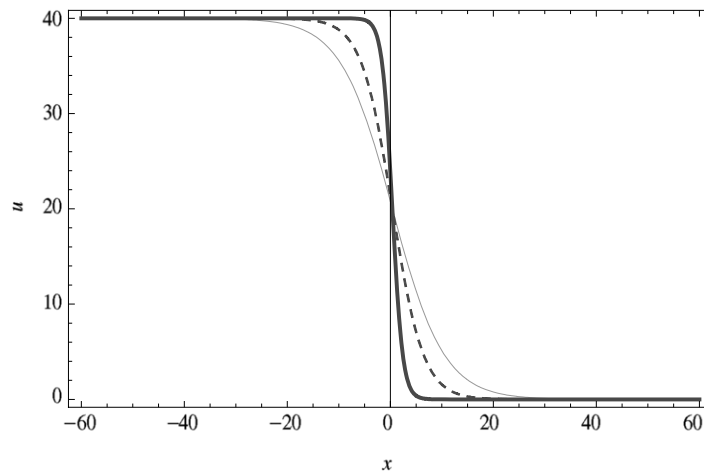


Fig. 1. The solution of Eq. (2) for different values of τ . The thin, dashed and thick curves represent the solutions for $\tau = 0, 10, 20$ respectively at $s = 1, p = 0.01, \omega = 0.2, \lambda = 10, \mu = 1, t = 2$.

By using (18), expression (14) can be written as

$$v(\xi) = \pm \frac{(s+1)\omega}{p\sqrt{\lambda^2 - 4\mu}} \left(\frac{G'}{G} \right) \pm \frac{(s+1)\lambda\omega}{2p\sqrt{\lambda^2 - 4\mu}} + \frac{(s+1)\omega}{2p}, \quad (19)$$

where $\xi = \pm \frac{s\omega}{(\tau\omega^2 - 1)\sqrt{\lambda^2 - 4\mu}}(x - \omega t)$.

Substituting the general solutions of (10) in (19), we obtain three types of travelling wave solutions of the time-delayed Burgers equation (Eq. (2)). We get the following hyperbolic functions solution when $\lambda^2 - 4\mu > 0$

$$u_1(x, t) = \left[\pm \frac{(s+1)\omega}{2p} \left(\frac{C_1 \sinh \frac{\sqrt{\lambda^2 - 4\mu}}{2} \xi + C_2 \cosh \frac{\sqrt{\lambda^2 - 4\mu}}{2} \xi}{C_1 \cosh \frac{\sqrt{\lambda^2 - 4\mu}}{2} \xi + C_2 \sinh \frac{\sqrt{\lambda^2 - 4\mu}}{2} \xi} \right) + \frac{(s+1)\omega}{2p} \right]^{\frac{1}{s}}, \quad (20)$$

where $\xi = \pm \frac{s\omega}{(\tau\omega^2 - 1)\sqrt{\lambda^2 - 4\mu}}(x - \omega t)$, C_1 and C_2 are arbitrary constants.

In particular, if $C_1 \neq 0, C_2 = 0$, then $u_1 = u_1(\xi)$ can be written as the following kink-like solution

$$u_1(x, t) = \left[\pm \frac{(s+1)\omega}{2p} \tanh \frac{\sqrt{\lambda^2 - 4\mu}}{2} \xi + \frac{(s+1)\omega}{2p} \right]^{\frac{1}{s}} \quad (21)$$

where $\xi = \pm \frac{s\omega}{(\tau\omega^2 - 1)\sqrt{\lambda^2 - 4\mu}}(x - \omega t)$.

In particular, if $C_1 = 0, C_2 \neq 0$, then $u_1 = u_1(\xi)$ can be written as the following soliton solution

$$\hat{u}(x, t) = \left[\pm \frac{(s+1)\omega}{2p} \coth \frac{\sqrt{\lambda^2 - 4\mu}}{2} \xi + \frac{(s+1)\omega}{2p} \right]^{\frac{1}{s}} \quad (22)$$

where $\xi = \pm \frac{s\omega}{(\tau\omega^2 - 1)\sqrt{\lambda^2 - 4\mu}}(x - \omega t)$.

The solution (22) is identical to those obtained in [1]. Next we obtain the following triangular periodic solution when $\lambda^2 - 4\mu < 0$

$$u_2(x, t) = \left[\mp \frac{(s+1)\omega}{2p} i \left(\frac{-C_1 \sin \frac{\sqrt{4\mu - \lambda^2}}{2} \xi + C_2 \cos \frac{\sqrt{4\mu - \lambda^2}}{2} \xi}{C_1 \cos \frac{\sqrt{4\mu - \lambda^2}}{2} \xi + C_2 \sin \frac{\sqrt{4\mu - \lambda^2}}{2} \xi} \right) + \frac{(s+1)\omega}{2p} \right]^{\frac{1}{s}}, \quad (23)$$

where $\xi = \pm \frac{s\omega}{(\tau\omega^2 - 1)\sqrt{\lambda^2 - 4\mu}}(x - \omega t)$, C_1 and C_2 are arbitrary constants.

It is clear that when $\tau = 0$, Eq. (2) reduces to the classical Burgers equation [1]. Taking the limit when $\tau \rightarrow 0$ in (21) and (22), we can recover exact particular solutions for the classical Burgers equation as in [1]. The effect of the time-delay of solution (21) is shown in Fig. 1.

4. Solutions to the time-delayed Burgers–Fisher equation

In this section, we obtain more travelling solutions of the time-delayed Burgers–Fisher equation. When $p = s = q = 1$ Eq. (1) reduces to the form

$$\tau u_{tt} + (1 - \tau)u_t + 2\tau uu_t - u_{xx} + uu_x - u + u^2 = 0, \quad (24)$$

where τ is a time-delayed constant. This equation may be called as time-delayed Burgers–Fisher equation. This equation shows a prototypical model for describing the interaction between the reaction mechanism, convection effect and diffusion transport [2,24]. It is clear that when $\tau = 0$, Eq. (24) reduces to the classical Burgers–Fisher equation discussed in [2,24]. After the customary transformation $u = v(\xi)$, $\eta = k(x - \omega t)$, we obtain

$$k^2(\tau\omega^2 - 1)v'' - k\omega(1 - \tau)v' - (2k\omega\tau - k)vv' - v + v^2 = 0. \quad (25)$$

To look for the travelling wave solutions of Eq. (24), we balance v'' and vv' in (25), which gives $m + 2 = 2m + 1$ so that $m = 1$. Now we can write (5) as

$$v(\xi) = a_1 \left(\frac{G'}{G} \right) + a_0, \quad a_1 \neq 0. \quad (26)$$

By substituting (15)–(17) and (26) in (25) and collecting all terms with the same power of $\left(\frac{G'}{G} \right)$ together, the left-hand side of (25) is converted into another polynomial in $\left(\frac{G'}{G} \right)$. Equating each coefficient of this polynomial to zero, yields a set of simultaneous algebraic equations for a_2 , a_1 , a_0 , ω and k . Solving the system of algebraic equations using *Maple*, gives the following set of nontrivial solutions

$$a_1 = \pm \frac{1}{\sqrt{\lambda^2 - 4\mu}}, \quad a_0 = \pm \frac{\lambda}{2\sqrt{\lambda^2 - 4\mu}} + \frac{1}{2}, \quad k = \pm \frac{1 + \tau}{\tau - 4} \frac{2}{\sqrt{\lambda^2 - 4\mu}}, \quad \omega = \frac{5}{2(1 + \tau)} \quad (27)$$

where λ , μ , δ and a_0 are arbitrary constants.

By using (27), expression (26) can be written as

$$v(\xi) = \pm \frac{1}{\sqrt{\lambda^2 - 4\mu}} \left(\frac{G'}{G} \right) \pm \frac{\lambda}{2\sqrt{\lambda^2 - 4\mu}} + \frac{1}{2}, \quad (28)$$

where $\xi = \pm \frac{1+\tau}{\tau-4} \frac{2}{\sqrt{\lambda^2-4\mu}} (x - \frac{5}{2(1+\tau)}t)$.

Substituting the general solutions of (6) in (28), we obtain three types of travelling wave solutions of the time-delayed Burgers–Fishers equation (Eq. (25)). We get the following hyperbolic functions solution when $\lambda^2 - 4\mu > 0$

$$u_1(x, t) = \pm \frac{1}{2} \left(\frac{C_1 \sinh \frac{\sqrt{\lambda^2 - 4\mu}}{2} \xi + C_2 \cosh \frac{\sqrt{\lambda^2 - 4\mu}}{2} \xi}{C_1 \cosh \frac{\sqrt{\lambda^2 - 4\mu}}{2} \xi + C_2 \sinh \frac{\sqrt{\lambda^2 - 4\mu}}{2} \xi} \right) + \frac{1}{2}, \quad (29)$$

where $\xi = \pm \frac{1+\tau}{\tau-4} \frac{2}{\sqrt{\lambda^2-4\mu}} (x - \frac{5}{2(1+\tau)}t)$.

In particular, if $C_1 \neq 0$, $C_2 = 0$, then $u_1 = u_1(\xi)$ can be written as the following soliton solution

$$u_1(x, t) = \pm \frac{1}{2} \tanh \frac{\sqrt{\lambda^2 - 4\mu}}{2} \xi + \frac{1}{2} \quad (30)$$

where $\xi = \pm \frac{1+\tau}{\tau-4} \frac{2}{\sqrt{\lambda^2-4\mu}} (x - \frac{5}{2(1+\tau)}t)$.

Next we obtain the following triangular periodic solution when $\lambda^2 - 4\mu < 0$

$$u_2(x, t) = \mp \frac{1}{2} i \left(\frac{-C_1 \sin \frac{\sqrt{4\mu - \lambda^2}}{2} \xi + C_2 \cos \frac{\sqrt{4\mu - \lambda^2}}{2} \xi}{C_1 \cos \frac{\sqrt{4\mu - \lambda^2}}{2} \xi + C_2 \sin \frac{\sqrt{4\mu - \lambda^2}}{2} \xi} \right) + \frac{1}{2}, \quad (31)$$

where $\xi = \pm \frac{1+\tau}{\tau-4} \frac{2}{\sqrt{\lambda^2-4\mu}} (x - \frac{5}{2(1+\tau)}t)$. The effect of the time-delay of solution (30) is shown in Fig. 2. And we found that the time-delay is effective in smoothing out the shock-wave nature of the travelling wave. When $\tau = 0$, Eq. (24) reduces to the classical Burgers–Fisher equation [24]. Taking the limit when $\tau \rightarrow 0$ in (30), we can recover an exact particular solution for the classical Burgers–Fisher equation as in [25].

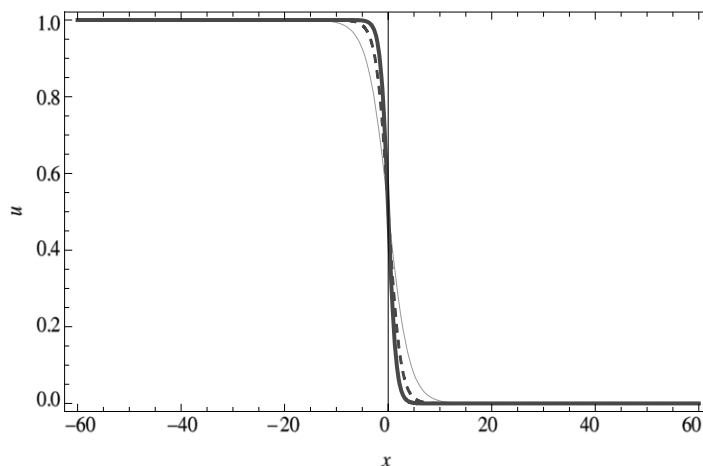


Fig. 2. The solution of Eq. (30) for different values of τ . The thin, dashed and thick curves represent the solutions for $\tau = 0, 0.5, 1$ respectively at $\lambda = 10, \mu = 1, t = 0.01$.

5. Conclusion

In this paper, $(\frac{G'}{G})$ -expansion method is implemented to obtain more exact solutions for time-delayed Burgers and time-delayed Burgers–Fisher equations. All solutions presented in this paper have been verified by using Mathematica by putting them back into the original equation. Finally, it is worth mentioning that the implementation of these proposed methods is very simple and straightforward, and it can also be applied to many other nonlinear time-delayed evolution equations.

References

- [1] E.S. Fahmy, H.A. Abdusalam, K.R. Raslan, On the solutions of the time-delayed Burgers equation, *Nonlinear Anal.* 69 (2008) 4775–4786.
- [2] E.S. Fahmy, Travelling wave solutions for some time-delayed equations through factorizations, *Chaos Solitons Fractals* 38 (2008) 1209–1216.
- [3] H. Jafari, V. Daftardar-Gejji, Solving a system of nonlinear fractional differential equations using adomian decomposition, *J. Comput. Appl. Math.* 196 (2006) 644–651.
- [4] X.Z. Li, M.L. Wang, A sub-ODE method for finding exact solutions of a generalized KdV–mKdV equation with high-order nonlinear terms, *Phys. Lett. A* 361 (2007) 115–118.
- [5] M.L. Wang, X.Z. Li, J.L. Zhang, Sub-ODE method and solitary wave solutions for higher order nonlinear Schrödinger equation, *Phys. Lett. A* 363 (2007) 96–101.
- [6] M.L. Wang, X.Z. Li, Extended F -expansion method and periodic wave solutions for the generalized Zakharov equations, *Phys. Lett. A* 343 (2005) 48–54.
- [7] M.L. Wang, X.Z. Li, Application of F -expansion to periodic wave solutions for a new Hamiltonian amplitude equation, *Chaos Solitons Fractals* 24 (2005) 1257–1268.
- [8] J.H. He, Homotopy perturbation method for bifurcation of nonlinear problems, *Int. J. Nonlinear Sci. Numer. Simul.* 6 (2005) 207–209.
- [9] T. Ozis, A. Yildirim, Traveling wave solution of Korteweg–de Vries equation using He's homotopy perturbation method, *Int. J. Nonlinear Sci. Numer. Simul.* 8 (2007) 239–243.
- [10] J.H. He, L.N. Zhang, Generalized solitary solution and compacton-like solution of the Jaulent–Miodek equations using the exp-function method, *Phys. Lett. A* 372 (2008) 1044–1047.
- [11] J.H. He, X.H. Wu, Exp-function method for nonlinear wave equations, *Chaos Solitons Fractals* 30 (1999) 700–708.
- [12] R. Sakthivel, C. Chun, New solitary wave solutions of some nonlinear evolution equations with distinct physical structures, *Rep. Math. Phys.* 62 (2008) 389–398.
- [13] V.A. Vladimirov, E.V. Kutafina, Exact travelling wave solutions of some nonlinear evolutionary equations, *Rep. Math. Phys.* 54 (2004) 261–271.
- [14] V.A. Vladimirov, C. Maczka, Exact solutions of generalized Burgers equation, describing travelling fronts and their interaction, *Rep. Math. Phys.* 60 (2007) 317–328.
- [15] E.G. Fan, J. Zhang, Applications of the Jacobi elliptic function method to special-type nonlinear equations, *Phys. Lett. A* 305 (2002) 384–392.
- [16] M.L. Wang, X. Li, J. Zhang, The $(\frac{G'}{G})$ -expansion method and travelling wave solutions of nonlinear evolution equations in mathematical physics, *Phys. Lett. A* 372 (2008) 417–423.
- [17] A. Bekir, Application of the $(\frac{G'}{G})$ -expansion method for nonlinear evolution equations, *Phys. Lett. A* 372 (2008) 3400–3406.
- [18] A. Bekir, A.C. Cevikel, New exact travelling wave solutions of nonlinear physical models, *Chaos Solitons Fractals* 41 (2009) 1733–1739.
- [19] S. Zhang, L. Dong, J.M. Ba, Y.N. Sun, The $(\frac{G'}{G})$ -expansion method for nonlinear differential–difference equations, *Phys. Lett. A* 373 (2009) 905–910.
- [20] S. Zhang, J.L. Tong, W. Wang, A generalized $(\frac{G'}{G})$ -expansion method for the mKdV equation with variable coefficients, *Phys. Lett. A* 372 (2008) 2254–2257.
- [21] S. Rendine, A. Piazza, L.L. Cavalli-Sforza, Simulation and separation by principle components of multiple demic expansions in Europe, *Am. Nat.* 128 (1986) 681–706.
- [22] A.J. Ammerman, L.L. Cavalli-Sforza, *The Neolithic Transition and the Genetics of Population in Europe*, Princeton University Press, 1984.
- [23] P.K. Galenko, D.A. Danilov, Selection of the dynamically stable regime of rapid solidification front motion in an isothermal binary alloy, *J. Cryst. Growth* 216 (2000) 512–536.
- [24] W. Xinyi, L. Yuekai, Exact solutions of the extended Burgers–Fisher equation, *Chinese Phys. Lett.* 7 (1990) 145–147.
- [25] E. Babolian, J. Saeidian, Analytic approximate solutions to Burgers, Fisher, Huxley equations and two combined forms of these equations, *Commun. Nonlinear Sci. Numer. Simul.* 14 (2009) 1984–1992.